

Tutorial Quiz 11 Rubric

Q1] Find z such that $z^2 = 3+4i$

if $z = a+bi$

then $z^2 = (a^2 - b^2) + 2abi$

equating the real & imaginary parts:

$$a^2 - b^2 = 3 \quad \text{①}$$

$$2ab = 4 \Rightarrow ab = 2 \quad \text{②}$$

Consider $b = 2/a$, where $a \neq 0$

Substituting in ①

$$a^2 - \left(\frac{2}{a}\right)^2 = 3$$

$$\Rightarrow a^2 - \frac{4}{a^2} = 3$$

$$\Rightarrow a^4 - 4 = 3a^2$$

on solving $a^2 = 4$ or -1

Since $a^2 > 0$

$$\therefore a^2 = 4$$

$$\therefore a = \pm 2$$

Substituting back in ②

$$\text{if } a = 2$$

$$b = 1$$

$$\text{if } a = -2$$

$$b = -1$$

$$\therefore z = 2 + i \quad \text{or} \quad -2 - i \quad \text{---(1)}$$

* Give mark for writing both or even one solution.

→ This question can also be solved using

De Moivre's Theorem (n^{th} root of a complex number). This approach is also valid. Even if final solⁿ is not exactly calculated.

$$\text{Find } z \text{ s.t. } z^2 = 3 + 4i$$

$$\text{Calc. modulus, } r = |z| = \sqrt{3^2 + 4^2} = 5$$

$$\text{Calc. argument, } \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Now, de moivre's theorem is:

$$\sqrt[n]{z} = \sqrt[n]{r} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right)$$

$$\text{for } k = 0, 1, \dots, n-1$$

here $n = 2$, $\therefore k = 0, 1$

$$\therefore \sqrt[2]{3+4i} = \sqrt[2]{5} \left(\cos \left(\frac{\tan^{-1}(4/3) + 2\pi k}{2} \right) + i \sin \left(\frac{\tan^{-1}(4/3) + 2\pi k}{2} \right) \right)$$

for $k = 0$

$$\sqrt[2]{3+4i} = \sqrt[2]{5} \left[\cos \left(\frac{\tan^{-1}(4/3)}{2} \right) + i \sin \left(\frac{\tan^{-1}(4/3)}{2} \right) \right]$$

for $k = 1$

$$\sqrt[2]{3+4i} = \sqrt[2]{5} \left[\cos \left(\frac{\tan^{-1}(4/3) + \pi}{2} \right) + i \sin \left(\frac{\tan^{-1}(4/3) + \pi}{2} \right) \right] \quad \text{--- (1)}$$

* Some students may write $\tan^{-1}(4/3)$ as approximately 53° . Which is also correct

Q2] a) Show u is harmonic

Computing the derivatives

$$u_x = -\sin x \cdot \cosh y$$

$$u_{xx} = -\cos x \cdot \cosh y$$

$$u_y = \cos x \cdot \sinh y$$

$$u_{yy} = \cos x \cdot \cosh y$$

$$\text{Since } u_{xx} + u_{yy} = -\cos x \cosh y + \cos x \cosh y = 0$$

→ (1)

$\therefore u$ is harmonic

b)

Cauchy - Riemann for $f = u + iv$

$$u_x = v_y \Rightarrow v_y = -\sin x \cdot \cosh y$$

$$u_y = -v_x \Rightarrow v_x = -\cos x \cdot \sinh y$$

Integrate v_x w.r.t. x

$$\rightarrow v(x, y) = -\sin x \cdot \sinh y + g(y)$$

Differentiate this w.r.t. y

$$v_y = -\sin x \cdot \cosh y + g'(y)$$

$$\text{But } v_y = -\sin x \cdot \cosh y$$

$$\therefore g'(y) = 0$$

$$\therefore g(y) = C$$

$$\therefore v(x, y) = -\sin x \cdot \sinh y + C$$

$$\therefore f(z) = u + iv$$

$$= \cos x \cdot \cosh y - i(\sin x \cdot \sinh y) + iC$$

$- (1)$